SURFACE ENGINEERING — MID-SEM QUICK NOTES

1. SURFACE ENGINEERING

Surface = outermost region interacting with environment. Surface atoms have fewer neighbours → unsatisfied bonds → higher energy (bulk: many neighbours, lower energy). Surfaces are energetically unfavourable (+ve free energy of formation).

Structure covers: atomic arrangement, planes, coordination no., surface energy, defects, rearrangement. Affects friction, wear, corrosion, fatigue, adsorption, wettability.

TLK model (Terrace–Ledge–Kink): **Terrace** = flat region (most coordinated); **Step/ledge** = atomically higher/lower (intermediate); **Kink** = irregular site at a step (most unsaturated bonds → most active). Sites differ in adsorption, diffusion, reaction, crystal growth.

Coordination no. (CN) = nearest neighbours. FCC bulk CN = 12; surface atom lower → more unsatisfied bonds → higher surface energy.

Planes: different planes → different surface energies. FCC: **(111) = lowest surface energy**.

Surface energy = excess energy per unit area. **Reduced by:** (1) reducing surface area, (2) exposing low-energy planes, (3) relaxation, (4) reconstruction, (5) adsorption of low-energy species, (6) coatings. *(2025: factors affecting surface energy)*

Relaxation: adjustment of spacing near surface; perpendicular spacing changes, parallel periodicity & symmetry unchanged; 1st layer moves slightly toward 2nd. Small displacement.

Reconstruction: significant rearrangement; surface atomic arrangement/periodicity changes; reduces surface energy through atomic rearrangement *(MCQ)*.

Adsorption: attachment/accumulation of atoms/molecules at surface (due to unsatisfied bonds); alters surface energy, wettability, friction, corrosion, reactions.

CORE CHAIN: surface atom → fewer neighbours → unsatisfied bonds → higher surface energy → seeks lower energy → relaxation / reconstruction / adsorption.

2. SURFACE-DEPENDENT PROPERTIES

Physical: roughness, colour, wettability. **Mechanical:** fatigue, wear, hardness, flexural/rupture strength. **Chemical:** corrosion, high-temp oxidation.

Roughness = unevenness; influences lustre, friction coeff., wear, corrosion resistance. High: more peaks/valleys → higher friction, wear, stress concentration, easier crack initiation. Low: better contact, fewer asperity effects.

Surface texturing = **preferred/controlled topography** (not random roughness) *(2025 MCQ)*.

Colour: how surface interacts with light; depends on condition, structure, finish, films/coatings.

Hardness = resistance to localized deformation/penetration; depends on microstructure & composition; affects friction, wear, some corrosion; improved by surface modification/treatment/coating.

Flexural strength = stress at failure in bending (≈ or slightly > tensile failure stress); affected by surface imperfections, microstructure, composition.

Links: roughness → stress conc. → crack initiation → fatigue; roughness → asperity contact → friction → wear; surface condition → environment → corrosion.

3. FAILURE ANALYSIS

Systematic investigation of failed component: **what → how → why failed → prevent recurrence.**

Genesis of failure: (1) **Design** – bad geometry, stress conc., wrong assumptions; (2) **Manufacturing** – defects, improper processing, surface damage; (3) **Service** – overload, corrosion, wear, fatigue, temperature; (4) **Material** – wrong material, microstructure problems, impurities.

Steps: collect history/evidence → visual exam → fracture surface exam → identify failure mode → NDT / chemical analysis / mechanical testing → root cause → corrective action → report.

Fracture types: Ductile – large plastic deformation, rough/fibrous, high energy. **Brittle** – little plasticity, sudden, flat/cleavage. **Fatigue** – initiation, propagation, final fracture; beach marks, striations. **Intergranular** – crack along grain boundaries.

NDT (no destruction): visual, ultrasonic, radiographic, magnetic particle, dye penetrant → locate cracks, defects, discontinuities, internal damage. **Chemical analysis:** composition, contamination, corrosion products, unexpected elements, material identity. **Mechanical testing:** hardness, strength, toughness.

4. FATIGUE (2025 Mid-Sem topic)

Fatigue = failure from repeated/fluctuating stresses; can occur **below static strength**.

Cyclic stress: $\sigma_m = (\sigma_{max}+\sigma_{min})/2$; $\sigma_a = (\sigma_{max}-\sigma_{min})/2$; $\sigma_r = \sigma_{max}-\sigma_{min}$; $R = \sigma_{min}/\sigma_{max}$

Mechanism: Stage 1 initiation – at surface roughness, scratches, notches, defects, inclusions, stress conc. **Stage 2 propagation** – $da/dN = A(\Delta K)^m$ (a = crack length, N = cycles, ΔK = stress intensity range = $K_{max}-K_{min}$, A, m = material constants). **Stage 3 final fracture** – remaining section can't carry load → sudden.

Why surface-dependent: cracks initiate at/near surface: rough surface → stress conc. → local high stress → crack initiation → failure. Surface residual stress also matters.

Characteristics: cyclic loading, initiation, propagation, final fracture, beach marks, striations, little overall plastic deformation vs ductile overload.

S–N curve: stress amplitude vs cycles to failure; amplitude ↓ → cycles ↑. **Fatigue limit** = stress below which failure may not occur in the test range.

Improve fatigue life: proper design, proper surface treatment, shot peening, laser shock peening, low plasticity burnishing, deep cold rolling → better surface condition + lower stress conc. + **compressive residual stress**.

5. WEAR

Wear = progressive material loss from surface due to relative motion with contacting substance(s). **Friction** = resistance to relative movement (contact + motion); can contribute to wear.

Classes: abrasive, adhesive, erosion, fatigue wear. Also: fretting, seizure, galling, oxidation wear, pitting, spalling, impact wear, brinelling.

Abrasive: hard particles/protuberances forced against & moving along a surface → ploughing/scratching → removal. **Low-stress:** scratching without severe crushing (chutes, sandy soils, ash & mineral handling). **High-stress:** severe interaction/crushing (mineral milling, earth-moving, loaded metal-metal sliding). Low vs high distinguished by **magnitude of applied stress** *(2025 MCQ)*. **Gouging:** very severe, large/hard particles, impact + ploughing → large grooves (mineral crusher). Rate ↑ with sharper, harder

abrasive, higher load, longer sliding; ↓ with harder surface; microstructure matters.

Adhesive: localized bonding between contacting surfaces, then transfer/loss. **ASPERITY BONDING** (asperity contact → local bond → motion → bond breaks → transfer) *(2025 tested)*.

Erosive: mechanical interaction with fluid / liquid / solid particles / multicomponent fluid; particles in flow impact surface. **Fatigue wear:** repeated rolling/sliding → cyclic stress → crack → fragment breaks away (pitting).

Fretting: very small-amplitude oscillatory motion + contact pressure → debris, damage, oxidation; with corrosion = **fretting corrosion**. **Seizure:** severe adhesion, surfaces bond, movement stops. **Galling:** severe adhesive wear – transfer, roughening, possible seizure. **Oxidation wear:** oxidation → oxide removed → fresh surface → repeat. **Pitting:** localized surface fatigue pits (cam paths, gear teeth, rails, metal tires). **Spalling:** large piece/layer breaks away (fatigue). **Impact wear:** repeated impact → deformation/removal. **Brinelling:** localized permanent indentation from concentrated contact load (bearings).

Factors: *Material* – hardness, strength, ductility, work hardening, microstructure. *Surface* – roughness, surface hardness, lubrication, opposing surface. *External* – load, speed, temperature, corrosion, opposing material.

MAP: hard particle → abrasion; asperity bonding → adhesive; particle/fluid impact → erosion; repeated rolling/sliding → fatigue wear; tiny oscillations → fretting; severe adhesion → seizure/galling; localized fatigue pits → pitting.

6. CORROSION

Corrosion = chemical/electrochemical degradation by environment. **Driving force:** system moves to lower free energy; higher-energy metal → lower-energy corrosion products.

Galvanic: 2 dissimilar metals + electrical connection + electrolyte; one becomes anodic & corrodes. Factors: potential difference, environment, polarization, area, distance, geometry.

Crevice: localized, in narrow gaps, joints, under deposits, shielded regions; crevice chemistry differs from exposed surface.

Pitting: protective film breaks locally → small anodic area → rapid dissolution → deeper pit. Highly localized, rapid, deep; chlorides & stagnant conditions promote (ship hull in sea – 2025).

Erosion-corrosion: erosion + electrochemical corrosion; flow/particles remove passive films & products → fresh metal → accelerated corrosion.

Cavitation: vapour bubbles form & collapse → shock/pressure → damage → film disruption → accelerated corrosion.

Selective leaching: preferential removal of one element; **brass: Zn removed = dezincification** (Zn in Cu alloys – 2025).

Hydrogen embrittlement: hydrogen interaction → loss of ductility, cracking. **SCC:** slow crack growth from synergy of mechanical stress + corrosion + susceptible material. **LME:** liquid metal + susceptible solid → ductility loss/cracking (Zn–Ga, Al–Ga, Cu–Hg, steel–Al/Pb/Hg).

High-temp oxidation: metal + O_2 at high T: $nM + (k/2)O_2 \rightarrow M_nO_k$. Topics: oxide layers, kinetics, alloy oxidation, oxide structure, internal oxidation.

MAP: dissimilar metals → galvanic; narrow gap → crevice; small hole → pitting; flow + corrosion → erosion-corrosion; bubble collapse → cavitation; element removed → leaching; hydrogen → embrittlement; stress + corrosion → SCC; liquid + solid metal → LME; high T + O_2 → HT oxidation.

7. MECHANICAL SURFACE TREATMENT

Shot peening (SP): bombard surface with small shots (steel, glass, ceramic beads) → indentations, cold work → plastic deformation → **compressive residual stress** → harder crack initiation → better fatigue. Cracks grow under tension, so surface compression counteracts tensile stress.

Coverage = fraction of surface impacted in a given time (%); shows indentation overlap & residual-stress uniformity. Higher/uniform → more uniform treatment; excessive/poorly controlled peening harms surface. **Avrami eq.:** C(t) with r = mean indentation radius, A = shot spread area, t = time, m = shot mass flow rate, ρ = shot density – *learn exact form from Lec 15–16 slide.*

Ultrasonic SP: high-frequency vibration drives media; ≈ 20 kHz, amplitude 50–200 μm.

Laser shock peening (LSP): intense pulsed laser > ~1 GW/cm² → surface vaporizes → high-T, high-P plasma → shock wave → plastic deformation → **deep** compressive stress. Less cold work than SP; depth ≈ 4× SP. Cons: expensive, controlled environment, skilled operators.

Low plasticity burnishing (LPB): smooth free-rolling spherical ball pressed with controlled normal force → near-surface plastic deformation → compressive stress → better fatigue; less cold work, can retard existing cracks.

Deep rolling: ball/roller pressed → high Hertzian compressive stress → yield exceeded → plastic deformation spreads as tool moves → smoother surface + compressive residual stress → improved fatigue.

MASTER CHAIN

SURFACE → fewer neighbours → unsatisfied bonds → surface energy → condition changes → roughness/defects → stress conc. → **fatigue crack initiation** → propagation → failure. **Surface + relative motion → WEAR. Surface + environment → CORROSION. Surface + mechanical treatment → compressive residual stress → better fatigue.**

Equations:

$\sigma_m = (\sigma_{max}+\sigma_{min})/2 \quad \sigma_a = (\sigma_{max}-\sigma_{min})/2$

$R = \sigma_{min}/\sigma_{max} \quad \sigma_r = \sigma_{max}-\sigma_{min}$

$da/dN = A(\Delta K)^m \quad \Delta K = K_{max}-K_{min}$

$nM + (k/2)O_2 \rightarrow M_nO_k$